Analysis I - 6c expanded solution

Product rule: $(uv)' = u'v + uv'$, Chain rule: $\dfrac{d}{dx}e^{g(x)} = g'(x)e^{g(x)}$.

$$
\begin{aligned}
f(x,y) &= (x+4y)\Big(e^{-2x}+e^{-3y}\Big),\\
\frac{\partial f}{\partial x} &= \underbrace{\frac{\partial}{\partial x}(x+4y)}_{=1}\cdot\Big(e^{-2x}+e^{-3y}\Big) + (x+4y)\cdot\underbrace{\frac{\partial}{\partial x}\Big(e^{-2x}+e^{-3y}\Big)}_{(-2)e^{-2x}+0}\\
&= \underbrace{e^{-2x}}_{\text{from }(x+4y)'} + e^{-3y} + \underbrace{(-2x)\,e^{-2x}}_{\text{from }x} + \underbrace{(-8y)\,e^{-2x}}_{\text{from }4y}\\
&= (1-2x-8y)\,e^{-2x}+e^{-3y}.
\end{aligned}
$$

Alternatively:

$$
f(x,y) = (x+4y)\Big(e^{-2x}+e^{-3y}\Big)
$$

Expand first, then differentiate term–by–term (treat y as a constant):

$$
\begin{aligned}
f(x,y) &= xe^{-2x}+xe^{-3y}+4ye^{-2x}+4ye^{-3y}.\\
\frac{\partial f}{\partial x} &= \frac{\partial}{\partial x}\Big(xe^{-2x}\Big)+\frac{\partial}{\partial x}\Big(xe^{-3y}\Big)+\frac{\partial}{\partial x}\Big(4ye^{-2x}\Big)+\frac{\partial}{\partial x}\Big(4ye^{-3y}\Big)\\
&= \underbrace{\Big(1\Big)e^{-2x}}_{\text{from }(x)'} + \underbrace{x(-2)e^{-2x}}_{\text{from }(e^{-2x})'} + \underbrace{e^{-3y}}_{\text{since }\frac{d}{dx}x=1} + \underbrace{4y(-2)e^{-2x}}_{\text{const. }4y} + \underbrace{0}_{\text{const. term}}\\
&= \Big(1-2x-8y\Big)e^{-2x}+e^{-3y}.
\end{aligned}
$$